Question 1

a). $17 - 12 + 13 - 18 = 5 + (-5) = 0$

b). $5 + 4 - 8 - 1 + 4 - 2 = 9 - 9 + 2 = 2$

Question 2

a). $\frac{4}{9} \div \frac{2}{9} = \frac{4}{9} \times \frac{9}{2} = \frac{4\times 9}{2\times 9} = \frac{4}{2} = 2.$

b). $\frac{2}{3} \div \frac{5}{4} = \frac{2}{3} \times \frac{4}{5} = \frac{2\times 4}{3\times 5} = \frac{8}{15}.$

Question 3

a). $2 \div \frac{1}{3} = 2 \times 3 = 6$

b). $\frac{1}{6} \div 2 = \frac{1}{6} \times \frac{1}{2} = \frac{1}{6\times 2} = \frac{1}{2}$

c). $3 \times \frac{-3}{4} = \frac{3\times(-3}{4} = -\frac{9}{4}.$

Question 4

a). $(-2)^3 \div 9 = -8 \div 9 = -\frac{8}{9}.$

b). $(-3)^2 = (-3) \times (-3) = +9.$

Question 5

a). $2\ 350\ 000 = 2,35 \times 10^6.$

b). $103\ 258\ 202 = 1,032\ 582\ 02 \times 10^8.$

c). $954\ 316\ 372,135 = 9,543\ 163\ 721\ 35 \times 10^8.$

d). $3\ 298,204 = 3,298\ 204 \times 10^3.$

Question 6

a). Non car $91 = 7 \times 13$.

b). Oui, car $\sqrt{97} \simeq 9,\cdots$ et 97 est impair (il finit par 7), il n'est pas multiple de 3 (car $9 + 7 = 16$ non plus), ni de 5 (il ne finit pas par 0 ou 5) ni de 7 ($97 \div 7 = 13$ reste 6), et le nombre premier suivant serait $11 > \sqrt{97}$.

Question 7

a). $250 = 2 \times 125 = 2 \times 5 \times 25 = 2 \times 5 \times 5 \times 5 = 2 \times 5^3.$

b). $108 = 2 \times 54 = 2^2 \times 27 = 2^2 \times 3 \times 9 = 2^2 \times 3 \times 3 \times 3 = 2^2 \times 3^3.$

Question 8

a). Pour $a = -2$, $5a + 1 = 5 \times (-2) + 1 = -10 + 1 = -9.$

b). Pour $a = -2$, $7a - 4 = 7 \times (-2) - 4 = -14 - 4 = -18.$

c). Pour $a = -3$, $7a - 4 = 7 \times (-3) - 4 = -21 - 4 = -25.$

Question 9

a). $(-3ac)(-2a^2) = (-3) \times (-2)aa^2c = 6a^3x$.

b). $(7ab)(-2a^3) = -14a^4b$.

c). $(-abc) \times (2bc) = -2ab^2c^2$

Question 10

a). $3a + 4b + 12b - 19a = (3 - 19)a + (4 + 12)b = -16a + 16b$.

b). $-8a + b - b^2 + a = -7a + b - b^2$.

c). $12ac + 41a - 32ca + 12c - 4a = 41a - 4a + 12ac - 32ac + 12c = 37a - 20ac + 12c$.

d). $7x^2 + 4x - 3 - 12x + 53 + 12x^2 - 1 = (7 + 12)x^2 + (4 - 12)x + 53 - 3 - 1 = 19x^2 - 8x + 49$.

Question 11

a). $-(-x + 7) = -(-x) - (+7) = +x - 7$

b). $-(-3x - 3) = +3x + 3$.

c). $-(x + 1) = -x - 1$.

Question 12

a). $-3(x+5)-4(x+2) = -3\times x+(-3)\times 5+(-4)\times x+(-4)\times 2 = -3x-15-4x-8 = -7x-23$.

b). $x + 4(2 - x) + 2 = x + 4 \times 2 + 4 \times (-x) + 2 = x + 8 - 4x + 2 = -3x + 10$.

c). $(x + 4) - 5(x + 1) = x + 4 - (5x + 5) = x + 4 - 5x - 5 = -4x - 1$.

Question 13

a).

$$\begin{aligned}
&7(x + 4) - (3x + 1) + x(x + 2) \\
&= 7 \times x + 7 \times 4 - 3x - 1 + x \times x + x \times 2 \\
&= 7x + 28 - 3x - 1 + x^2 + 2x \\
&= x^2 + 7x + 2x - 3x + 28 - 1 \\
&= x^2 + 6x + 27
\end{aligned}$$

b).

$$\begin{aligned}
-(x + 1)x - x - 1 &= -(x \times x + 1 \times x) - x - 1 \\
&= -(x^2 + x) - x - 1 \\
&= -x^2 - x - x - 1 \\
&= -x^2 - 2x - 1
\end{aligned}$$

c).

$$\begin{aligned}
x + x(1 - x) - 2x(x + 1) &= x + x \times 1 + x \times (-x) - (2x \times x + 2x \times 1) \\
&= x + x - x^2 - (2x^2 + 2x) \\
&= 2x - x^2 - 2x^2 - 2x \\
&= -3x^2
\end{aligned}$$

Question 14

a). $15ac + 3ab + 6a = 3a \times (5c) + 3ab + 3a \times 2 = 3a(5c + b + 2)$.

b). $3x + xy - 5xz = x(3 + y - 5z)$.

c). $4xy + 12x = 4x(y + 3)$.

d). $10xy + 20yz = 10y(x + 2z)$.

Question 15

a). $(x + 1)(x + 2) + (x + 2)(x + 3) = (x + 2)[(x + 1) + (x + 3)] = (x + 2)(2x + 4)$

b). $2(x + 1) + 3(x + 1) + x(x + 1) = [2 + 3 + x](x + 1) = (5 + x)(x + 1)$.

c). $(x+1)(x+3) + (x+3)(2x-1) = (x+3)[(x+1) + (2x-1)] = (x+3)[3x] = 3x(x+3)$.

Question 16

a).

$$\begin{aligned}(-x-5)(12x+1) &= (-x) \times (12x) + (-x) \times 1 + (-5) \times (12x) + (-5) \times 1 \\ &= -12x^2 - x - 60x - 5 \\ &= -12x^2 - 61x - 5\end{aligned}$$

b).

$$\begin{aligned}(2x-3y)(4x-2) &= 2x \times 4x + 2x \times (-2) + (-3y) \times 4x + (-3y) \times (-2) \\ &= 8x^2 - 4x - 12xy + 6y\end{aligned}$$

c).

$$\begin{aligned}(2a+3b)(-4a+6b) &= (2a) \times (-4a) + (2a) \times (6b) + (3b) \times (-4a) + (3b) \times (6b) \\ &= -8a^2 + 12ab - 12ab + 18b^2 \\ &= -8a^2 + 18b^2\end{aligned}$$

d).

$$\begin{aligned}(x+2)(2x+2) &= x \times 2x + x \times 2 + 2 \times 2x + 2 \times 2 \\ &= 2x^2 + 2x + 4x + 4 \\ &= 2x^2 + 6x + 4\end{aligned}$$

Question 17

a).

$$\begin{aligned}(-3xy-2y)(x+5) &= (-3xy) \times x + (-3xy) \times 5 + (-2y) \times x + (-2y) \times 5 \\ &= -3x^2y - 15xy - 2xy - 10y \\ &= -3x^2y - 17xy - 10y\end{aligned}$$

b).

$$\begin{aligned}(-4x+1)(y-3) &= -4x \times y + (-4x) \times (-3) + 1 \times y + 1 \times (-3) \\ &= -4xy + 12x + y - 3\end{aligned}$$

Question 18

a). On écrit successivement

$$\begin{aligned} 4x+7 &= 51 \\ 4x+7-7 &= 51-7 \\ 4x &= 44 \\ \frac{4x}{4} &= \frac{44}{4} \\ x &= 11 \end{aligned}$$

On vérifie aussi que $4 \times 11 + 7 = 44 + 7 = 51$, donc 11 est l'unique solution.

b). On écrit successivement

$$\begin{aligned} 7x+8 &= 43 \\ 7x+8-8 &= 43-8 \\ 7x &= 35 \\ \frac{7x}{7} &= \frac{35}{7} \\ x &= 5 \end{aligned}$$

On vérifie aussi que $7 \times 5 + 8 = 35 + 8 = 43$ donc 5 est l'unique solution.

c). On écrit successivement

$$\begin{aligned} 2x-3 &= 19 \\ 2x-3+3 &= 19+3 \\ 2x &= 22 \\ \frac{2x}{2} &= \frac{22}{2} \\ x &= 11 \end{aligned}$$

On vérifie aussi que $2 \times 11 - 3 = 22 - 3 = 19$ donc 11 est l'unique solution.

d). On écrit successivement

$$\begin{aligned} 5x-23 &= 77 \\ 5x-23+23 &= 77+23 \\ 5x &= 100 \\ \frac{5x}{5} &= \frac{100}{5} \\ x &= 20 \end{aligned}$$

On vérifie aussi que $5 \times 20 - 23 = 100 - 23 = 77$ donc 20 est l'unique solution.

Question 19

a). On a $1\ cm^3 = 0,001\ dm^3$ donc $50\ cm^3 = 0,05\ dm^3$

b). On a $1\ dm^3 = 1\ 000\ cm^3$ donc $40\ dm^3 = 40\ 000\ cm^3$.

c). On a $1\ m^3 = 1\ 000\ dm^3$ donc $0,3\ m^3 = 300\ dm^3$.

Question 20

a). Le volume est $(3\ cm) \times (5\ cm) \times (15\ cm) = 225\ cm^3 = 225\ mL$.

b). Le volume est $(3\ cm) \times (5\ cm) \times (4\ cm) = 60\ cm^3 = 60\ mL$.